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21 January 1983
SUPERSEDING
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# MILITARY SPECIFICATION

## TRUCKS, FORK, REACHING AND TIERING, NARROW AISLE, ELECTRIC, 3000 POUNDS CAPACITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers standup rider, end control battery powered, industrial, telescoping mast, reaching and tiering trucks, the load capacity of which is rated at 24 inches from the front face of the forks.

1.2 Classification. Trucks shall be of the following type, styles, and sizes, as specified: (See 6.2).

Type E -- Non spark-enclosed.
Type EE -- Spark-enclosed.
Style I -- Outriggers, pallet straddling.
Style II -- Outriggers, non pallet straddling.
Size 1 -- 144 inches lift, 96 inches overall height.
Size 2 -- 130 inches lift, 89 inches overall height.
Size 3 -- 180 inches lift, 96 inches overall height.
Size 4 -- 198 inches lift, 89 inches overall height.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Navy Ships Parts Control Center, Post Office Box 2020, Attention: Code 0302, Mechanicsburg, Pa. 17055 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3930

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SPECIFICATIONS

Military

MIL-P-514 — Plate, Identification Instruction and Marking, Blank.

MIL-F-3541 -- Fittings, Lubrication (Hydraulic).

MIL-G-3859 — Grease Gun, Hand, High Pressure, Lever-Operated Cartridge and Bulk Loading (14 Ounce and 21 Ounce Capacity).

MIL-M-3971 — Meter, Time Totalizing, Operational Use, Electric.

MIL-L-4387 — Lubrication Dispensing Equipment Accessories.

STANDARDS

Federal

FED-STD-595 — Colors.

W-B-00133 -- Battery, Storage (Lead-Acid Industrial For Cycle Service).

Military

MIL-STD-105 -- Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 -- Marking for Shipment and Storage.

MIL-STD-130 -- Identification, Marking of U. S. Military Property.

MIL-STD-162 — Preparation for Delivery of Warehouse Materials Handling Equipment for Domestic and Overseas Shipment and Storage.

MIL-STD-461 — Electromagnetic Interference Characteristics Requirements for Equipment.

MIL-STD-642 -- Identification Marking of Combat and Tactical Transport Vehicles.

MS-15367-------Battery,Storage, For Electrical Powered Industrial Trucks and Tractors.

MS-16955-------Hook Type Fork Mounting For Trucks of 2000 and 4000 Pounds capacity, Based on 24 Inch Load Center.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)



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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply. Where there is a conflict between this specification and any referenced publication, this specification shall govern.

UNDERWRITERS' LABORATORIES, INC.
PUBLICATIONS

Standard for Electric-Battery-Powered Industrial Trucks (UL 583).

(Application for copies shall be addressed to: Underwriters' Laboratories, Inc. 1285 Walt Whitman Road, Melville, Long Island, New York 11746; 207 East Ohio Street, Chicago, Illinois 60611; or 1655 Scott Boulevard, Santa Clara, California 95050.)

AMERICAN NATIONAL STANDARDS INSTITUTE
Safety Standard for Powered Industrial Trucks (ANSI B56.1)

(Application for copies should be addressed to: The American Society of Mechanical Engineers, 345 East 47th Street, New York, New York 10017)

DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION
29 CFR, Cahpter XVII, Part 1910.
Occupational Safety and Health Standards.
(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402.)

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION
NEMA Standards

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street N.W., Washington, D.C. 20037)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC.
SAE Handbook

(Application for copies should be addressed to: Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, Pennsylvania 15096).

3. REQUIREMENTS

3.1 Standard product. The trucks covered by this specification shall be complete and ready for operation when delivered. Batteries shall be as specified in 3.2.2.1. Trucks shall be constructed and equipped to meet all requirements specified herein. The end item and major and minor assemblies shall be new and of a standard design or model product used by the end item

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manufacturer current at time of bidding, except for such minor modifications from manufacturer's standard as may be required to conform to these specification requirements. The component parts of the unit need not be the products of the same manufacturer. However, the manufacturer shall be so established in the industry that prompt and continuing service and delivery of repair parts will be assured.

3.1.1 Preproduction model. Unless otherwise specified (see 6.2), the suppliers shall furnish a truck for examination and test within the time frame specified (see 6.2), to prove prior to starting production that his production methods and choices of design detail will produce trucks that comply with the requirements of this specification. Examination and tests shall be as specified in section 4, and, unless otherwise specified herein, shall be subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examination and tests as specified. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer shall specify which examination and tests will be conducted by the Government and, which examination and tests will be conducted by the supplier. (see 6.2). The approved preproduction model may be presented as the last unit for delivery under the contract after being rehabilitated to appear new. This shall include the furnishing of new tires.

3.1.2 Material. Material not definitely specified shall be at least equal to the end item manufacturer's normal commercial product, and shall meet the test requirements specified in section 4. The Government reserves the right to require any bidder who proposes to furnish other than a "normal commercial product" or government specified item to submit, prior to and as a condition precedent to the award of contract, detailed engineering drawings of those parts he proposes to furnish in lieu thereof for approval of the contracting officer.

3.1.3 Design. Design of trucks covered by this specification and all components therein, shall conform to accepted design practices and shall insure safe, efficient, and economical operation under normal service conditions. Where used in this specification, "easily accessible" shall refer to the truck after the battery is removed. There shall be no evidence of accelerated wear or failures or permanent deformation when operated under normal service conditions and when tested as specified herein.

3.1.3.1 Safety. Trucks shall conform to the applicable requirements of American National Standards Institute publications: Standard for Electric-Battery-Powered Industrial Trucks ANSI B56.3 and Safety Standard for Powered Industrial Trucks ANSI B56.1. Trucks shall also comply with the requirements of OSHA 29CFR, Chapter XVII, Part 1910.

3.1.3.1.1 Certification. The contractor shall submit to the contracting officer or his authorized representive satisfactory evidence that the trucks furnished under this specification meet the requirements of 3.1.3.1. Acceptable evidence of meeting these requirements shall be a UL label, or a certified test report from a recognized independent testing laboratory acceptable to the government, indicating that the trucks conform to the requirements of 3.1.3.1.

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3.1.3.2 Capacity. The load capacity of the truck shall be not less than 3,000 pounds rated at 24-inch load center from the vertical face of the forks at maximum fork height with the reach mechanism fully extended.

3.1.4 Certificates of compliance. When specified herein, or upon request, a certificate of compliance with the requirements of this specification shall be furnished. Such certificates shall be supported by back-up data which shall be made available for review. The certificates shall read as follows:

This certifies that the . . . . . (Component, part or assembly) complies with all the provisions and requirements of Specification . . . . . paragraph . . . . . The certificate shall be signed by a duly authorized agent of the contractor.

3.1.5 Electrical equipment.

3.1.5.1 Resistors. Resistors shall not be used in series with the motor armature for speed control. Speed shall be controlled by means of a solid-state control system as specified in 3.2.1.3. Low resistance shunts across field coils may be used to increase top speed of the motor. Such resistors as are necessary in the control circuit network shall be continuous-duty types and of a power rating sufficient to withstand all tests required by this specification.

3.1.5.2 Overcurrent devices. As required by UL 583, overcurrent devices shall be provided to protect the power circuits from damage resulting from internal or external wiring defects or other faulty operation. The overcurrent device(s) shall not function while performing test operations specified herein, including ramp climbing.

3.1.5.3 Control Panel. The control panel shall be designed for easy removal as a unit or for front removal of individual major components mounted thereon. The panel cover shall be readily removable without interference from the truck structure or other major components. Bolts, mounting studs, and nuts shall be recessed in back of panel and shall be secured so that they will not turn or become loose when the electrical assembly is removed or replaced on the panel. Leads to and from the panels or components shall, to the maximum extent practicable, be harnessed in compact groupings. All leads and harnesses shall be of sufficient length for ease of removal of panel or components mounted thereon.

3.1.5.4 Conductors. Conductors shall be located within the truck frame and body, in conjunction with suitably designed noncombustible shielding, to provide mechanical protection from damage. All wires and cables shall be color or number coded for identification. All wiring external to the truck or when maneuvering the truck shall be encased in rigid metal conduit, except that flexible metal conduit may be used at flexure areas; similar protection shall be provided for exposed wiring runs in operator's platform area. Conductors passing through compartment walls or other metal barriers shall be protected by loom, rubber bushings, or other protective material and shall be securely anchored to prevent movement of conductors relative to the opening. All terminal connectors shall be fastened with spring lock washers, lock nuts, or other means of preventing accidental loosening as a result of vibration or unintentional movement of wire terminal ends. Cables to the remote control shall be of heavy duty, oil resistant type and protected from damage by

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location and position regardless of lift. Suitable means shall be provided to guide these cables within the truck body so accidental scraping against walls and other objects will not damage the cables.

3.1.6 Initial Production. When specified (see 6.2), the supplier shall furnish to the Government one or more trucks for testing as specified in 4.4.

3.2 Power System.

3.2.1 Motor and motor accessories.

3.2.1.1 Motors. Drive motors shall be either series or compound wound, heavy duty, industrial type, with a high starting torque, designed for efficient operation at 24 volts direct current. Motors shall be capable of withstanding the mechanical stresses and current loads induced by tests as required in section 4. The resultant strains and temperature rise shall not cause failure of any component part, excessive pitting, or other than normal blackening of the commutators. The hydraulic pump motor(s) shall be series or compound wound. The hydraulic pump motor(s) shall have sufficient capacity to lift rated load at speeds specified herein, in addition to extension and retraction requirements. Motor(s) shall be directly connected to the pump. Motors shall be capable to withstanding sustained operation without evidence of insulation breakdown, short circuit, or failure of any part or component.

3.2.1.2 Motors - Type EE Trucks. In addition to the requirements of 3.1.5.2. motors(s) for type EE trucks shall be provided with one or more thermal relay switches which shall cause disconnection of battery power to the affected circuit before the outside surface temperature of the motor(s) exceed limits as specified in UL 583. Thermal relays shall be located in critical heat areas and shall interrupt the current only to the over-heated circuit.

3.2.1.2 Insulation. Insulation for motors shall be the class A, B, F or H or combination thereof as defined in NEMA standards. Insulation shall be selected to give the best balance of overall operating characteristics of torque, speed, and temperature rise. The class of insultation shall be determined by the material with the lowest temperature rise used in the insulation system. Armature and fields need not have the same class of insulation, but each part shall meet the temperature rise permitted for its class of insulation. The electrical and mechanical properties of the insulated winding shall not be impaired by the application of the temperature permitted (the work "impaired" is here used in the sense of causing any change which could disqualify the insulating materials for continuous service).

3.2.1.3 Speed controller. Speed and directional control(s) shall be provided by a vertical or rotary control(s) mounted for right hand operation. Control(s) shall automatically return to a neutral position under any of the following conditions: (a) release of the control, (b) Release of deadman brake pedal, or (c) when operator dismounts truck. The speed controller shall provide infinitely variable truck speed up to 75 - 85% of the maximum truck speed. Control and power circuits shall incorporate solid-state components that will insure operations as specified herein and minimize power losses through devices as a consequence of voltage drops through the rectifying or control elements. Control system design shall insure continuous-duty operation

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without excessive heating of the motors due to eddy-currents or hysteresis losses caused by current pulsations or so-called "spikes" in the applied voltage when tested as specified herein.

3.2.1.4 Contactor(s) assembly. Contactors shall be the magnetic clapper or internal solenoid type, designed for efficient operation at 24 volts. Contactors on a given truck shall be designed for maximum interchangeability of parts with easily replaceable tips of silver or copper alloy composition and shall be manufacturer's standard component. All contractors shall be readily accessible.

3.2.2 Battery and battery accessories.

3.2.2.1 Battery. Unless otherwise specified (see 6.2), the supplier shall furnish batteries for each truck. Unless otherwise specified (see 6.2), batteries shall conform to W-B-133C and MS-15367-8.

3.2.2.1.1 Battery connectors. Trucks shall be equipped with a 2-contact type EC standard battery connector or type SB connector with adapter furnished for connecting with mating parts of type EC connectors. Charging connectors or receptacles shall be so arranged that they do not permit movement of the vehicle under its own power while connected to the charging source.

3.2.2.1.2 Emergency battery disconnect. A disconnect device, included with the battery connector, shall be provided to disconnect all power from the battery. The device shall be conveniently located so that it can be actuated by the operator without leaving the compartment.

3.2.2.1.3 Type EE trucks. In order to insure that no current is flowing when the connector is being disconnected on type EE trucks a magnetic contactor shall be installed in one of the two main power lines from the battery connector. The control circuit for the magnetic contactor shall be wired with two switches in series, either of which shall cause the contactor to open and disconnect all power. The first switch shall be a seat switch. The second switch shall be an emergency switch with a positively attached mushroom type button located within easy reach of the operator.

3.2.2.2 Battery compartment. Battery compartment or support area shall accomodate the battrey specified with clearance on all sides conforming to UL 583 requirements. Effective means shall be provided to restrain the battery from moving more than a total of 1/2 inch in a horizontal direction. The compartment shall be constructed to permit easy removal or servicing of the battery such as checking water level and refilling. When compartment construction only permits side removal of the battery, rollers or other means shall be provided to facilitate battery removal or replacement by a force not exceeding 75 pounds to maintain movement after battery is in motion. Both sides of the compartment shall be hinged or easily detachable with hand-operable quick opening fasteners to permit battery removal, inspection, and servicing. The inside of the compartment shall be protected with an acid resistant paint to resist deterioration through normal contact with electrolyte. Compartment shall be easily accessible for cleaning and battery servicing and shall contain no inaccessible pockets or reservoirs which are difficult to keep clean.

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3.2.2.3 Battery protector. A battery discharge indicator shall be provided. The indicator shall be a solid state device requiring no adjustment except a perodic light bulb check or replacement. It shall be tamper proof and carry a UL label. It shall be designed to protect the battery from approaching a dangerously low state of discharge by first warning the operator by means of a signal light, or fuel gage and signal light. After a short interval of time or after consumption of an additional amount of electrical energy, it shall interrupt the lift control circuit making the lift mechanism inoperative until the battery has been replaced or recharged. The circuit shall be such that the operator cannot activate the lift control circuit by unplugging and replugging the battery after it has been 80% discharged, (typically 1150 specific gravity). Unplugging and replugging the battery prior to reaching 80% discharge shall not reactivate the lift control circuit. The discharge indicator shall reset when a battery charged to 56% of full capacity(typically 1200 specific gravity) is put on the truck. The discharge indicator may have controls to adjust the discharge setting and the reset level to values other than 80% and 56%, provided such settings can be made without electrical test equipment and the positions of the settings are indicated by indelible markings on the controls. When a gage is used, it shall be mounted on the driver panel, and it shall display energy in the familiar empty to full type display used in gasoline vechiles. There shall be a defined point on the gage where the warning light goes on and a defined point where the truck lift mechanism becomes inoperative. Location and method of mounting shall provide plain visibility of the indicator to the driver without creating interference with truck operation, maintenance, and entry and egress movements of the operator.

3.2.3 Drive unit. The all gear drive shall be a reduction unit(s) directly connected to the drive axles. The motor, reduction unit, and drive wheel(s) shall be integrally turret mounted and shall revolve on large diameter hardened ground bearings. Gear units, shall be totally enclosed and operate in a constant oil bath. Final reduction may be in a totally enclosed casing or be arranged for positive periodic lubrication of final drive gearing to wheel. Drive gear housings shall be equipped with filler and magnetic drain plugs readily accessible for servicing without removing other components. With drive wheels locked, the drive assembly shall withstand without failure, fracture, or deformation of any component a static test torque applied to the input of the transmission with test load to the calculated as follows:

$$T = \frac{0.04\ WD}{R}$$

where, T = test torque in foot-pounds
W = combined weight of truck and load in pounds
D = diameter of drive wheels in inches
R = overall gear ratio of the drive system.

When specified (see 6.2), a certificate of compliance shall be provided in lieu of test. When test is specified, the complete drive assembly may be tested prior to or after complete assembly of the truck at the option of the contractor.

3.2.4 Hydraulic system. The hydraulic system shall consist of all the hydraulic components necessary for operation of the forks and mast. The power used for the system shall be furnished by a hydraulic pump(s) directly driven

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by an electric motor(s) independent of the traction motor. The lifting system shall provide for smooth lowering of load at a speed not greater than 80 feet per minute in event of failure or damage to hydraulic lines. Provision shall be made to protect hydraulic system against excessive pressures due to overloading. All hydraulic lines shall be routed to provide the shortest line, minimum number of bends, ease of maintenance and maximum protection. Provisions shall be made to prevent damage of lines due to chafing. Sump tank and hydraulic system shall be capable of being drained without spillage. The storage tank breather and filler cap, lift, tilt, and extension cylinders shall be provided with means to protect against contamination of hydraulic fluid by direct impingement or seepage of water. A filter type breather cap shall be provided. A removable strainer not coarser than 50 mesh, installed for ease of inspection and cleaning, shall be provided to prevent entry of foreign material into storage tank and the hydraulic system. A removable 25-micron full-flow filter with emergency bypass shall be provided to filter all the oil.

The storage tank shall be of welded steel construction and shall have a capacity not less than 110 percent of the amount of fluid required for the hydralic system's operation. In addition, it shall have sufficient capacity to prevent air from entering the hydraulic system with all hydraulic pistons fully extended and shall have sufficient free air capacity to prevent oil from being discharged through tank filler cap when maximum return flow of oil is surged into the tank from the system. Piston rods for hydraulic cylinders shall be provided with a hard chrome plate finish. Surfaces shall be thoroughly cleaned prior to plating in accordance with normal commercial practice. Plating shall be electro-deposited hard chromium finish, having uniform thickness of not less than 0.0005 inch. Lifting and other cylinders, if any, shall be provided with "wiper rings" to preclude entry of dust and dirt into the packing. Hydraulic valves and cylinders shall have means for preventing oil seepage from dripping or running onto electrical elements.

3.2.4.1 Hose reels. When hydraulic hose reels or other types of takeups are used on the mast to contain hydraulic hose, they shall be designed to facilitate containment, release and retraction of the hose without entanglement. Sheave wheels, guide rollers, guide blocks, or a mechanism in the hose reel may be used to maintain hose tension.

3.2.4.2 Hydraulic pressure hose. Hydraulic pressure hose shall conform to SAE J517, except 100R7 type hose pressure tubing shall be in accordance with SAE J524 or SAE J525, tube fittings shall conform to SAE J514 and SAE J518.

3.3 Structure.

3.3.1 Chassis and frame. The frame and related structure shall be fabricated from structural steel and steel castings at least equal to the size and design of the manufacturer's commercial frame members. The frame and related structure shall be sufficiently strong and well braced to prevent racking or twisting that might result in permanent distortion of any component part or member. The frame and related structure shall have sufficient depth to afford protection for working parts. It shall be capable of withstanding the frame racking test specified in 4.6.2.11 without evidence of permanent deflection or distortion. Unless otherwise specified (see 6.2), the test shall be performed, except when previously conducted by the manufacturer. In this case, certified test data shall be made available.

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3.3.2 Body. The truck body shall be constructed of heavy sheet steel or plate equal to that used in the manufacturer's commercial product. In addition, the truck body shall provide protection for the drive unit assembly, battery assembly, steering mechanism, and other parts requiring protection. Provision shall be made for immediate and easy access to all working parts requiring inspection and maintenance. Covers shall be secured with bolts or quick opening fasteners removable with ordinary hand tools. Overall truck width shall be such that the truck can perform all applicable tests specified herein.

3.3.2.1 Operator's platform. The operator's platform shall be fabricated from steel plate and shall be provided with non-slip rubber tread not less than 1/4 inch thick, cement bonded thereon. The operator's platform shall be to the right of the drive turret and shall have sufficient depth to permit the operator to stand within the plan outline of the truck and permit unobstructed egress from the rear of the truck. It shall be strong enough to prevent collapsing and pinching the operator in event of collision. The operator's compartment shall provide protective space with minimum dimensions of 14 inches front to back and a minimum of 13 1/2 inches width at the platform floor. Guards shall be constructed to withstand a compression load equal to the weight of the loaded vehicle applied in the direction of travel.

3.3.3 Uprights. Uprights shall be of the telescopic type with sufficient strength and rigidity to withstand 300-percent of rated load with forks retracted and 200-percent with forks extended without permanent deformation or failure. Cross members of the mast structure, when in a lowered position, shall not interfere with the operator's horizontal or downward line of sight directly to the front of the vehicle. The operator shall be able to see the tip of at least one fork at any lift height without leaving his operating position when there is no load on the forks. Mast rollers with anti-friction bearings shall be provided between telescopic sections and main frame upright. Adequate means by either replacement or adjustment shall be provided in the uprights to compensate for wear between lateral rolling contacts without disassembly of the mast. Upper and lowerlimit stops shall be provided to prevent overtravel. The upright lift arrangement shall be designed to control premature elevation of an intermediate section during raising or "hanging-up" of these sections during lowering.

3.3.3.1 Forks. Steel forks shall be provided which shall be capable of withstanding, without permanent deformation, a static test load of 300-percent of rated load as well as stresses induced by tests specified in section 4. Forks shall be attached in a manner which prevents lateral slippage or permanent deformation of lift carriage components. Forks shall be adjustable laterally without the need for hand tools. Fork thickness shall not exceed 1/2 inch at the tip, and the end taper shall be not less than 14 inches long. Taper shall be on the underside of the fork. Unless otherwise specified (see 6.2), forks shall have the following nominal dimensions and spacing adjustment measured between the outer edges of the forks. (See Table I.) Fork mounting shall conform to MS 16955.

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TABLE I

| | |
|---|---|
| Thickness (maximum) | 2 1/16 inches |
| Width (maximum) | 5 1/16 inches |
| Length | 40 + 0 - 1/2 inches |
| Spacing, minimum, outside (forks spread) | 30 inches |
| Spacing, maximum (forks together) | 14 inches |

3.3.3.2 Load back rest. A metal back rest removable without disturbing any other component shall be provided. It shall be 47 to 48 inches high measured from load carrying surfaces of the forks and shall be a minimum width at least equal to the width of the carriage. It shall have width, capacity and strength sufficient to prevent a rated load 48 inches high or less from forcing back against the uprights. The backrest shall be firmly attached to the lift carriage but shall be readily removable with common hand tools. The back rest shall retain its perpendicular relationship to the forks at all degrees of tilt. Backrest design shall not unduly obstruct operator's vision. The spacing of vertical members shall be no more than 6 inches between members. The back rest shall be capable of withstanding, without permanent deformation, a test load equal to the rated load uniformly distributed over the area of the back rest.

3.3.3.3 Fork carriage. The fork carriage shall provide lateral and longitudinal absorption of load thrusts without failure of any component when subjected to 300-percent of rated load. The carriage shall be mounted on the mast on adjustable rollers equipped with antifriction bearings.

3.3.4 Outriggers. Outriggers shall be fabricated from steel plate or bar steel formed to shape as an integral part of the frame to provide a rigid structure and shall be capable of withstanding without permanent deformation all stresses induced by tests specified in section 4.

3.3.4.1 Style I trucks. Inside spacing between outriggers shall be not less than 50 inches to permit straddling of a 40-inch by 48-inch pallet on the forks at floor level. Outriggers shall have an outside width not exceeding 60 inches.

3.3.4.2 Style II trucks. Outriggers for style II trucks shall have an outside width of 48 inches + 1/2 inch maximum.

3.3.3.4 Reach mechanism. A reach mechanism shall be provided which will permit extension of the forks or fork assembly to a point where the front vertical face of the forks is at least 3/4-inch in front of the forward edge of the load wheels, tires, or outriggers to permit handling of loads at floor level that are off center laterally or would otherwise conflict with the outriggers or load wheels. A chain, if used, for extensible reach mechanism shall be easily adjustable in an amount equal to not less than one complete link without removal of any truck structure or major component. The reach mechanism structure shall be capable of meeting the performance requirements specified herein and shall be sufficiently strong and well braced to prevent racking or twisting that might result in permanent distortion of any component part. The design must be particularly well suited to withstand stresses which result from bumping of one fork on one corner of the load or against an obstruction. Antifriction bearings, bronze, or steel bushing shall be provided for the rollers on the pantograph type extension and the extensible mast reach

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mechanism.

3.3.5 Wheels. Trucks shall be mounted on flat-faced steel or malleable iron wheels, equipped with one of the following configurations: (1) two independently suspended and driven drive wheels; (2) one drive wheel and one mechanically linked idler wheel; or (3) one drive wheel and one or more caster follower wheels. The non-drive wheels shall be loaded or the entire system designed for laterally articulated mounting to maintain drive wheel contact with uneven floors. Trucks shall be equipped with four tandem-mounted polyurethane load wheels mounted in the outriggers. Wheels, when equipped with new tires, shall clear any part of the truck structure by at least 1/4 inch under all conditions of operation. Readily accessible grease fittings or pre-lubricated sealed bearings shall be furnished to provide lubrication in accordance with good automotive practice. Load wheels on outriggers shall be provided with means to assure equal load distribution on each of the load wheels with and without the rated load on the truck.

3.3.5.1 Tires. The drive wheel tire shall be polyurethane or industrial solid rubber, smooth low profile type or wide lug cushion type, molded onto a steel band. Band shall be of the press-on type or shall have an integral flange for direct bolting onto a driving hub. Tires for caster wheels shall be either the press-on or mold-on type. Minimum tire size for the drive wheel shall be 10 inches diameter with 5 inches face. Minimum tire size for caster or idler wheel(s) shall be 6 inches diameter with 2 inches face for duals and 8 inches diameter with 3 inches face for singles. Load wheel tires shall be 5 inches nominal diameter by 2 5/8 inches minimum width.

3.3.6 Steering. Steering shall be accomplished by a horizontal steering wheel equipped with steering knob for convenient operation. The steering control shall have the hand wheel mounted with the upper face 40 to 45 inches above the operator's platform and in such a manner that will permit the operator to steer with his left hand when facing the load. The steering mechanism shall turn the drive wheel to provide for right angle stacking in one motion as specified herein. The steering mechanism shall turn the drive wheel through spur gear reduction or accepted commercial equivalent reduction device which shall permit ease of operation. A hydraulic steering booster or other power assisted steering shall be provided. Chains, if used in the steering mechanism, shall be easily adjustable in an amount equal to not less than one complete link without removal of any truck structure or major component. Clockwise or right hand rotation of the steering wheel shall provide for right hand or clockwise turning of the truck while in forward motion (load first). Provision shall be made for precision adjustment to compensate for wear of all major steering parts and for lubrication of all friction points by accessible grease fittings or a lubricant reservoir. The hydraulic steering or power assisted steering fluid reservoir shall be readily accessible for checking of lubricant level.

3.3.7 Brakes. Brakes shall be hydraulically activated and of the external contracting type, disc type, or internal expanding shoe type mounted on the traction motor shaft, countershaft, or drive wheel. Brake lining, when used, shall be of molded or woven type and shall be cement-bonded or riveted to the brake shoe. Brakes shall be operated by the operator's left foot and shall be automatically spring applied when the brake is released or when the operator leaves the truck. The braking system shall be electrically interlocked with a

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switch or switches which shall interrupt the electrical circuit to the traction motor when the brake is applied. The brake adjustment shall provide for at least 90 percent contact of brake lining with drum throughout service life of the lining. The operator's brake pedal shall be located within a clear space at least 6 inches wide. Pedal shall be at least 4 1/2 inches wide and provided with a rubber covering or other non-slip material to prevent slippage. The brake pedal, when depressed to fully release the brake, shall be in a horizontal plane, flush with the operator's platform.

3.4 Controls and instrumentation. Where applicable, the provisions of ANSI B56.3 shall apply. Controls and instrumentation shall be of the heavy duty type and shall be constructed to be moisture and weather resistant to prevent the entry of moisture when truck is operated or stored out of doors under all weather conditions.

3.4.1 Controls. All switch and lever type controls shall be provided with permanent position markings (decals are not acceptable) either at the switch or lever or, in the case of the directional speed controls, on a diagram visible to the operator. Distance from the near edge of one knob to the near edge of an adjacent knob or other surface shall be a minimum of 1 1/2 inches. All controls shall have a directional motion insofar as possible. That is, the motion of the hand control shall be similar to the resulting motion of the truck operation to be performed.

3.4.1.1 Directional speed controls. Selective forward and reverse speed control(s) shall be provided and shall be conveniently located for safe and efficient right hand operation with operator facing load. Controls shall also provide for safe, convenient, and efficient operation when operator faces away from load.

3.4.1.2 Lift, reach and tilt controls. Lift, reach and tilt controls shall be self-centering, conveniently located for right-hand operation. Controls shall be located such that lift control is at the extreme right, reach control at the next position, and tilt at the extreme left. The controls shall provide for lifting and lowering of the load when the truck is in motion without requiring the operator to release the steering control or utilize any part of his body other than hands to operate controls. Rearward or upward motion of lift, reach, and tilt control levers shall elevate, retract or tilt load rearward. Modulation of the speed of lift, or lowering, reach, and tilt shall be readily controllable. Lifting speed shall be infinitely variable from zero to the maximum lifting speed specified.

3.4.1.3 Accelerator control. A hand operated accelerator control shall be provided. A control button or grip may be combined with the directional control and shall be installed for comfortable right hand operation.

3.4.1.4 Extension-retraction control. The forks shall be extended by movement of the control from the center position to the forward position. Retraction of the forks shall be accomplished by movement of the control from the center position to the rearward position. Means shall be provided to prevent jerky stopping or starting of forks at the end of the extension and retraction movements.



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3.4.1.5 Remote control. When specified (see 6.2), each truck shall be equipped with separate controls in such a position that the operator, while standing on a safety pallet placed on the forks, shall be able to operate the elevating and lowering mechanism over its complete range. These controls shall be so located that when not in use they will not interfere with normal operation of the truck or constitute a safety hazard. These controls shall be protected so they cannot be operated or damaged by commodities falling on them. They shall be so connected that operation of the lift mechanism cannot be accomplished unless deadman brake is set and operation of the truck cannot be accomplished except from the remote control position while remote control is in use. Lift-lower controls shall be spring-loaded to return to the off position upon release. An automatic take-up devide shall be provided on the mast or truck body to eliminate slack in the remote control cable and shall be positioned to prevent interference with normal truck operation. Lift-lower controls shall be arranged for two hand operation so that operator must actuate separate controls with both hands simultaneously to cause carriage to elevate or lower in order to prevent any possible accidents to either hand or arm while carriage is elevating or lowering. Means shall be provided for the operator on the platform to shut off the power of the truck.

3.4.1.6 Key switch. A key operated switch and key shall be furnished (see 6.5). The key shall be removable only when the switch is in the "off" position. When the switch is in the "off" position, control circuits shall be open. All trucks procured under each contract shall be keyed identically.

3.4.2 Instruments. All gages and meters shall be products of manufacturers regularly engaged in producing these types of instruments. They shall be flush mounted in a location convenient to and plainly visible to an observer standing on the floor.

3.4.2.1 Hour meter. An electric operational hour meter which registers the total number of operating hours of the traction and pump motors only shall be provided. Hour meters shall conform to Specification MIL-M-3971, type I, grade B. Hour meter shall be wired to record only powered operations.

3.5 Accessories. Where applicable, the provisions of UL 583 shall apply. All accessories shall be constructed to resist moisture, rain and weather when the truck is operated or stored outdoors under all weather conditions.

3.5.1 Horn. Truck shall be equipped with an electric push button operated horn. Push button shall be mounted and locaed for convenient operation by the operator. The horn shall be operable without requiring the operator to release the steering or directional controls.

3.5.2 Operator's overhead safety guard. An overhead safety guard for the protection of the operator shall be provided in accordance with ANSI B56.1. Guard height, measured from ground level to top side of the guard frame members shall not exceed 96 inches for sizes 1 and 3 and 89 inches for size 2 and 4. The guard shall extend to within 2 1/2 inches of the rear plan outline of the truck. Guard design shall not interfere with operation of truck, or with normal movements of the operator when entering, leaving, or operating the truck. Unless otherwise specified (see 6.2) a welded fixed overhead guard shall be provided.

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3.5.3 Safety shield. A transparent tempered or laminated safety glass shield shall be mounted on the mast in such a manner as to prevent injury to the operator's hands or arms by the lift chains or cross members. The shield shall not interfere with the operational requirements of the truck or with the control knob clearances. It shall not obstruct the vision of the operator.

3.5.4 Spotlight. Each truck shall be equipped with one hand adjustable, sealed beam spotlight of not less than 25 watts, mounted on the mast and located for right hand operation. Directional focus of this light shall be vertically and horizontally adjustable. Adequate protection by position or guard shall be provided to prevent damage to the light under all operating conditions. An on-off switch shall be incorporated into the control panel. Spotlight shall be wired independent of other electrical circuits.

3.6 Performance

3.6.1 Lifting speed. Speed of lift with rated load (ambient temperature between 65 degrees and 85 degrees F) shall be not less than an average of 30 feet per minute from floor to maximum fork height and not less than 40 feet per minute without load under the same conditions, forks retracted in both instances. The truck shall be equipped with a freshly charged battery. The lifting system shall provide for precise positioning of unloaded forks.

3.6.2 Lowering speed. Speed of lowering the unloaded forks (ambient temperature between 65 degrees F and 85 degrees F), forks fully retracted, shall be not less than 40 feet per minute or more than 80 feet per minute. The hydraulic control system shall be damped, metered or easily manually regulated to reduce shock and prevent the truck from overturning when rated load is fully extended and lowered at maximum speed and stopped suddenly.

3.6.3 Extension and retraction. Truck shall be capable of extending rated load (ambient temperature between 65 degrees and 85 degrees F), at any height, a horizontal distance not less than the distance from the forward face of the forks (retracted) to at least 3/4 inch forward of outermost end of the outriggers or load wheels, whichever is further, in not more than 5 seconds.

3.6.4 Steering. The drive assembly shall be capable of being turned from extreme right to extreme left and vice versa in not more than 9 turns of the hand steering wheel with a maximum steering effort of 20 pounds when the truck is stationary on a smooth dry concrete floor with and without rated load in the carrying position, forks retracted. The steering gear assembly shall be capable of withstanding an unbalanced tangential force of 150 pounds applied to the steering wheel without permanently distorting or deforming any part of the steering mechanism.

3.6.5 Speed. When carrying a rated load on a level surface, trucks shall be capable of forward and rearward speeds of not less than 4 miles per hour when equipped with a fully charged battery.

3.6.6 Slope ascension, forward direction. The truck shall be capable of ascending a 10 percent grade on a dry concrete surface when carrying the rated load. Length of ramp shall be not less than 15 feet. The truck shall be able to accelerate from a dead stop on this slope when carrying the rated load.



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3.6.7 Acceleration. From a standing start, the truck carrying rated load shall have an average acceleration rate in the forward and reverse directions such as to negotiate a distance of 22 feet in not more than 7 seconds. There shall be not more than one second delay in movement of the truck when truck is accelerated from a stop.

3.6.8 Service brakes. The brakes shall enable the truck to develop a drawbar drag equal to at least 12-percent of the gross weight of the truck (including rated load) while truck is being towed in both forward and reverse directions over a level, smooth, dry concrete surface.

3.6.9 Load tilt. The tilt mechanism shall be capable of bringing rated load at least level when fully elevated and extended. It shall be capable of providing a positive rearward tilt of 4 degrees with rated load on the forks at a load carrying position of 17-24 inches above the floor with the forks retracted. All of the above measurements are based on the relationship of the top face of the forks with the truck on a level flat platform. Forward and rearward tilt shall be controllable at all fork heights or extensions.

3.6.10 Overall height. Overall height of unloaded truck, measured from the ground to the top of the highest part of the truck, shall not exceed 96 inches for sizes 1 and 3 and 89 inches for size 2 and 4.

3.6.11 Maximum fork height. Fork height of loaded truck, measured from the ground to the top face of the forks with forks horizontal, uprights vertical, and load extended shall not be less than 144 inches for size 1, 130 inches for size 2, 180 inches for size 3, and 198 inches for size 4.

3.6.12 Free lift height. Height of forks without exceeding specified collapsed mast height shall be not less than 40 inches with rated load on the forks.

3.6.13 Underclearance. The truck shall have sufficient underclearace to permit operation without load from one horizontal plane to another, up and down a 15 percent slope. The truck shall have sufficient underclearance to permit operation with load from one horizontal plane to another, up and down a 10-percent slope. With the forks flat on the operating surface the truck shall have adequate underclearance to permit operation.

3.6.14 Hydraulic system. The hydraulic system shall show no permanent deformation or failure or leakage when the truck is subjected to 300-percent or rated load, forks retracted, and 200-percent of rated load, forks extended. This requirement does not include the pump. The pressure relief valve(s) shall be set to open when pressure equals 120-percent of the maximum pressure required to perform any operation supplied by a given pump(s).

3.6.15 Endurance test. The truck shall be capable of undergoing the endurance test specified in 4.6.2.18 without deformation, breakdown, failure, or replacement of any component (except for the battery), or any other defect and with no change in performance requirements specified herein and with no appreciable signs of wear in either the steering mechanism, the lift and tilt assembly, extension-retraction assembly, or drive assembly. Wear in the steering mechanism is considered to be an increase in free play of wheel or more than 1 1/2 inches measured at the steering wheel outer periphery. Wear in



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the lift assembly is defined as any apparent inability to lift rated load to maximum lift height and hold for a period of not less than 5 minutes and with not more than one inch drift.

3.6.16 Maneuverability. Trucks shall be capable of right angle stacking in aisles as specified below with a 40-inch by 48-inch loaded pallet. The 48-inch dimension shall be across the forks. Style I, size 1 and 2 trucks shall be capable of removing or placing this loaded pallet in a stack with four (4) inches clearance on each side of the load, in 78-inch aisles and style I, size 3 and 4 trucks shall be capable of removing or placing this loaded pallet in a stack with four (4) inches clearance on each side of the load in 81 inch aisles. Each category shall require not more than 60 seconds for each removal or placement. Style II trucks shall also be capable of performing this requirement with only three (3) inches clearance on each side of the loaded pallet.

3.6.17 Stability. The truck shall conform to the stability requirements of ANSI B56.1 when tested as specified in 4.6.2.10.

3.6.18 Overhead guard strength. The driver's overhead safety guard shall be of sufficient strength to support a uniformly distributed static test load of 6,000 pounds. It shall also be capable of withstanding the impact of a solid 100-pound hardwood cube (or equivalent) dropped a distance of 5 feet ten times without fracture or permanent deflection exceeding 3/4 inch.

3.6.19 Power Consumption. The truck, when equipped with a fully charged battery as specified shall operate continuously for not less than 5 hours when tested as specified in 4.6.2.17.

3.6.20 Retraction overload. Forks, mast assembly and related structures shall show no permanent deformation or failure when subjected to a retraction overload of 300 percent of rated load when tested as specified in 4.6.2.9.2.

3.6.21 Extension overload. Forks, mast assembly and related structures shall show no permanent deformation or failure when subjected to an extension overload of 200 percent of rated load when tested as specified in 4.6.2.9.3.

3.7 Bearings. Rotating parts shall be mounted on ball or roller bearings to the extent normally provided in standard commerical practice and in accordance with the best automotive practice. Bearing load capacity, including applicable safety and life factors, as recommended by the bearing manufacturer, shall be equal to or greater than maximum bearing loading under normal truck operation.

3.8 Lubrication. All surfaces requiring lubrication shall be provided with suitable means for lubricating.

3.8.1 Lubricants. The truck shall operate as specified herein when lubricated with standard military lubricants (see 6.4).

3.8.2 Lubrication fittings. Lubrication fittings shall conform to SAE J534. Fittings shall be located in a protected position and shall be accessible to a hand-operated grease gun conforming to MIL-G-3859 with a 10-inch flexible extension. Accessibility to fittings shall be provided without the removal or

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adjustment of accessories or parts. Panel and plates equipped with hand-operable, quick-disconnect fasteners or no more than eight common fasteners may be removed to provide accessibility.

3.8.3 Lubrication seals. All lubrication seals shall be properly sized and seated to prevent leakage of lubricant through or around positioned seals.

3.8.4 Screw threads. Screw threads shall be in accordance with Bureau of Standards Handbook H28.

3.8.5 Pressure - release device. An automatic-release device shall be provided where the use of pressure lubricating equipment could damage grease seals or other parts.

3.9 Finish treatment and painting. All exterior surfaces of the truck shall be thoroughly cleaned and shall be dry and free from mill scale, oil, grease, dirt, and rust and shall be painted as soon as practicable after cleaning with not less than one coat of metalic base primer and two coats of synthetic enamel. Primer and finish coats shall be applied in accordance with normal commerical practice to assure complete coverage and durability of the finish. The finish color shall be No. 13538 (gloss yellow) in accordance with FED-STD-595A. The finish coat when dry shall be a smooth even surface, free from runs, sags, blisters, peels, chips, and areas of no film or thin film. Surfaces of components and assemblies of the truck not normally painted with a finish coat shall be cleaned and prepared in accordance with standard commercial practice for the forklift truck industry. Working surfaces (i.e. lift chains, sprockets, and rollers, etc.) plastics, instruments, and such other components not normally painted with the finish coat shall be adequately protected or masked from possible overspray.

3.10 Electromagnetic interference. When specified (see 6.2), truck shall be equipped for control of electromagnetic interference in accordance with MIL-STD-461B, Part 8, Class C1, Group II.

3.11 Identification marking. The truck shall be identified with an identification plate conforming to MIL-P-514, type I, style 1, Composition C, (GG-P-455 Type I, Grade A, Class 1) which shall be located in a visible and safe location on the instrument panel or firewall of the truck. The identifying data to be applied to the identification plates shall be as indicated herein and as specified in MIL-STD-130. Identification data shall include as a minimum contract number, serial number, model number, safety rating, empty weight, payload, location and capacity of lifting and tiedown points and center of gravity. The truck shall also be equipped with instruction, warning, and caution plates conforming to MIL-P-514, prominently located and describing any special or important procedure to be followed in operating, loading, and servicing of the truck or its components. All plates shall be securely attached to the truck with screws, bolts, or rivets and shall be furnished and mounted by the supplier.

3.11.1 Instruction plates. The forklift shall be equipped with instruction plates conforming to MIL-P-514, type III, composition C(GG-P-455, type I, grade A, class 1)material, including warnings, cautions and diagrams describing any special or important procedures to be followed in assembly, operating or servicing the forklift.

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3.12 Registration data and vehicle marking. The manufacturer shall execute registration data cards, as required by the contracting officer. Unless otherwise specified, the assigned registration number for each truck shall be painted by the contractor, in black enamel, in block numbers 3 inches high on each side of the upright and on each side of the vehicle mast. In addition, the gross vehicle weight (GVW) with the required battery shall be marked on each side of the truck. The letter shall be gothic capitals, and the numerals shall be 2 inches high where space permits. Trucks shall be marked in black color number 17038 in accordance with Federal Standard No. 595. Marking paint shall be in accordance with TI-E-489. Characters shall be block-type capitals and Arabic numerals. Marking location and information and character size shall be as follows:

| Information | Example | Location | Character Size |
|---|---|---|---|
| Capacity | 3,000 Lbs. Capacity | On each side of mast | 2 inch height |
| Rider warning | NO RIDERS | On rear of mast | 2 inch height |
| Service Designation | U.S. Navy | On each side and rear | 1-1/2 inch height |

3.13 Maintainability. Provision shall be made for adjustment, servicing and replacement of all electrical assemblies and components, battery, wearing parts of lift and tilt mechanism, brakes and components, wearing parts of steering assembly, tires, wheels, lights, and horn without the removal of any major parts other than access covers. If hand access openings are used on the exterior of the truck, the edges of each opening shall be smooth and shall be provided with a hinged cover, whenever it is required. Dimensions of access openings shall be in accordance with SAE J925 for a bare hand or normal clothed man. Drainage of lubricants shall be to the ground without draining on any part of the truck. When drains are located over other parts of truck, integral tubes, pipes, troughs or other means shall be used to convey the lubricant from the drain to the ground. Drains shall be accessible without the removal or adjustment of accessories or parts other than opening covers. Drain holes with removable drain plugs or covers shall provide for complete drainage of all reservoirs and driveline housings without disconnecting hose(s). The hydraulic system reservoir shall be equipped with a dipstick to determine the fluid level. As a minimum, the dipstick shall be graduated with two marks indicating 'full' and 'add'. The dipstick shall be accessible without the use of handtools. Oil filler caps shall be painted orange-yellow in accordance with FED. STD. 595, Color No. 13538. The truck shall not require scheduled maintenance at intervals of less than 1,000 circuits of the endurance test as specified in 4.6.2.18.

3.14 Interchangeability. Corresponding units and replaceable assemblies, subassemblies, and parts having the same part number shall be physically and functionally interchangeable as complete units without modification. Where dimensions, ratings, tolerances, and other characteristics are not specified, the manufacturer's design limits shall be used to determine compliance with the foregoing.



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3.15 Workmanship. The truck shall withstand any operation specified herein without permanent deformation, breakage of connections, malfunction, or component interference caused by improper workmanship. All parts of the truck before and after painting shall be clean and free from sand, rust, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. Edges and surfaces exposed to operating and maintenance personnel shall be smooth and rounded to the extent that a hazardous surface does not exist.

3.15.1 Steel and other metal fabrication. Metals used in the fabrication of the truck shall provide original quality surface finish and shall be free from kinks and sharp bends. Metals having corroded or pitted surfaces are not acceptable. The straightening of materials shall be done by methods that will not cause weakening or injury to the material. Burrs and sharp edges in holes and on sheets, plates, and members shall be removed sufficiently to assure correct fits and to prevent loosening of fasteners and damage to components. Flame cutting with a tip suitable for the thickness of metal may be employed instead of shearing or sawing. Splatter shall be removed from exposed cuts and from re-entrant cuts. Heated metals shall be allowed to cool slowly, except in the performance of designed heat treatment, and overheating shall be avoided in accordance with the recommendations of the metal manufacturer. All bends of a major character and all modular assembly fabrication shall be made with controlled means to insure uniformity of size and shape.

3.15.2 Welds. Welded joints shall be sound, smooth, free from pits, holes, fissures, and rough projecting edges and shall be thoroughly fused to the base metal.

3.15.3 Bolted connections. Bolt holes shall be accurately formed and shall have the burrs removed. Washers or lockwashers or a combination of the two shall be provided on all bolts, studs, and capscrews having straight thread. Selflocking nuts are acceptable in lieu of standard nuts and lockwashers. All fasteners shall be correctly tightened and shall have full engagement.

3.15.4 Castings. Castings shall be of the best quality normally used in commercial practice and shall be of uniform quality and condition, free from blowholes, porosity, hard spots, shrinkage defects, cracks, misplaced coring, or other injurious defects. Castings shall be commercially machinable and shall not be repaired, plugged, or welded to adversely affect the strength, use, or machinability. They shall be of such quality as to preclude weeping of internal fluids through the castings as a result of porosity and to permit machining for leak free sealing at cover and closure points. Castings shall be suitably spot faced or machined to assure proper seating of fasteners such as bolts, lockwashers, nut assemblies and cap screws.

3.15.5 Rivet connections. Rivet holes, in the sizes recommended in standard practice, shall be accurately punched or drilled and shall have burrs removed. Rivets shall be driven with power tools. Rivet head shall be full, neatly made, concentric with rivet holes and in full contact with the surface of the member. Excessive upsetting of rivets to fill holes will not be acceptable.

3.15.6 Machine work. All parts shall be manufactured to gage through the use of correct jigs, fixtures, or tape controlled machines or combination of these. Like parts shall be interchangeable.

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3.15.7 Transportability. Unless otherwise specified (see 6.2), the truck shall be transportable by highway, rail, and marine modes of transport and by C-130 and C-141 aircraft. Transportability approval for Army Equipment must be obtained from the Army Transportability Agent (see 4.10)

3.16 Static discharge device--type EE trucks. Trucks shall be equipped with two replaceable electrically conductive flexible straps or non-ferrous chains to discharge static electrically through contact with the road surface. The electrical resistance between the truck chassis and the floor through the statio discharge device shall not exceed 250,000 ohms.

3.17 Technical manuals. One copy of the technical manual conforming to MIL-M-21861 shall be furnished by the manufacturer with each truck at time of shipment. The manual shall be placed in a protected place in the equipment. Each manual shall be preserved and packaged in a bag of the transparent material conforming to Type II of MIL-F-22191 by Method IC-1 of MIL-P-116 and the bag heat sealed. Additional copies of the manual shall be furnished as specified.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection responsibility. Unless otherwise specified in the contract or order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. When performed, sampling inspection shall be in accordance with the requirements of MIL-STD-105.

4.2 Classification of inspections. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3)
(b) Initial production inspection (see 4.4)
(c) Quality conformance inspection (see 4.5)
(d) Inspection comparison (see 4.8)
(e) Inspection of preparation for delivery (see 4.9)

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction truck shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The preproduction truck shall be tested as specified in 4.6.2.1 through 4.6.2.20. Failure of any test shall be cause for rejection.

4.3.3 Post-test examination. Upon completion of all tests the truck shall be examined in accordance with 4.7. Evidence of any defects or excessive wear which cause failure during tests or might cause failure in subsequent operation

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of the truck shall be cause for rejection.

4.4 Initial production inspection. When specified (see 3.1.6), one or more initial production trucks will be selected at random by the Government from the trucks being produced by production tooling and will be examined as specified in 4.6.1 and tested as specified in 4.6.2.1 through 4.6.2.19 to determine conformance to the requirements of this specification. The inspection will be performend by the Government at a site selected by the Government. Acceptance of an initial production truck shall not exclude the remaining trucks from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any or all other tests contained in this specification as part of the initial production test, and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production truck to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production truck and shall be cause for refusal by the Government to continue acceptance of production trucks until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on trucks previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence from all trucks accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Examination. Each truck shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.5.2 Tests. Each truck shall be tested as specified in 4.6.2.1.4, 4.6.2.2, 4.6.2.5 and 4.6.2.6. All gauges, instruments, lights, horns, switches, controls and other operating components shall be tested to determine that they perform the function for which intended. Failure to meet any test requirement shall be cause for rejection.

4.6 Inspection procedure.

4.6.1 Examination. The truck shall be examined for completeness of assembly, presence of components, specified dimensions, specified materials, accessories, quality of workmanship, identification marking, registration data, provisions for repair and maintenance, interchangeability, resistance to moisture and weather, bearing and lubrication requirements, finish and compliance with safety standards.

4.6.2 Tests. Tests to determine compliance with the specified requirements shall be conducted as outlined herein.

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4.6.2.1 Lift and tilt assembly.

4.6.2.1.1 Fork height. Elevate rated load to maximum lift height with load extended. Measure the distance from the ground to the top face of the forks with load horizontal. The fork height shall not be less than specified in 3.6.11.

4.6.2.1.2 Free lift height. Elevate rated load (forks retracted) until the inner mast, fork carriage or forks begins to exceed the specified collapsed mast height. Measure the true vertical distance from floor to top surface of the forks at rated load center. The free lift height shall be as specified in 3.6.12.

4.6.2.1.3 Load tilt. The truck with rated load in carrying position (forks retracted) shall be placed on a level horizontal surface. With forks parallel to the floor, mark a vertical reference line on the side of load. Tilt load to maximum rearward position and measure angle of travel by referenced line. The load shall then be raised to maximum lift height, fully extended and maximum rearward tilt. Load must be at least parallel to the floor with no downward tilt. A clinometer, protractor or other measuring device may be used for this test. Load tilt as measured shall not be less than specified in 3.6.9.

4.6.2.1.4 Lifting speed and lowering speed. Starting with a fully charged battery and hydraulic oil at, ambient temperature, the empty forks, fully retracted, shall be raised to maximum height. Actuate lower control and record time required for unloaded forks to reach their fully lowered position. Repeat this operation ten times, calculate average time, and compute lowering speed. Lowering speed shall be as specified in 3.6.2. Lowering speed shall also be checked under simulated hydraulic line failure to determine compliance with 3.2.4. After computation of lowering speed, place rated load on the forks with center of gravity of the load positioned 24 inches from the front face of the forks. Raise load at maximum lifting speed to full lift height and record the time required to reach this height. Repeat lift cycle five times and calculate average time. Repeat test without load. Speed of lift with and without rated load shall not be less than specified in 3.6.1. The following equation may be used in computing lifting and lowering speeds:

$$\text{Speed of lift/lowering (ft./min.)} = \frac{\text{Full lift height (inches)} \times 5)}{\text{Average time (seconds)}}$$

4.6.2.2 Extension and retraction. Raise rated load to maximum lift height in five (5) interval steps. Extend and retract rated load at various heights to the full horizontal distance. Measure this distance and record the time in seconds from the instant extension or retraction begins to the end of extension or retraction to determine compliance with 3.6.3.

4.6.2.3 Underclearance test. The truck shall be operated from one horizontal plane up a 15 percent ramp to another horizontal plane and down the ramp without rated load. Observe whether all parts of the truck clear the surface. The truck, with rated load on the forks, shall be operated from one horizontal plane up a 10 percent ramp to another horizontal plane and down the ramp to determine compliance with 3.6.13.

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4.6.2.4 Steering.

4.6.2.4.1 Steering gear. The steering mechanism shall be tested to determine that the operator can comfortably turn the wheel, without interference from adjacent truck structure, from extreme left to extreme right and vice versa, within limits specified in 3.6.4. Forces shall be applied to the steering knob in the plane of the wheel. Test shall be conducted with truck stationary on dry level concrete.

4.6.2.4.2 Steering overload. The truck shall be tested to determine compliance with the steering gear assembly overload requirements of 3.6.4. Loads shall be applied in the plane of the steering wheel. The steering wheel shall be turned to full right and full left to determine compliance with this requirement. The test shall be conducted 5 times in each direction alternately with the steering mechanism at the extreme turn of the steering wheel to the right and to the left.

4.6.2.4.3 Free play. The steering mechanism shall be inspected on preproduction sample unit(s) to determine compliance with 3.6.15. Measurements shall be taken before and after endurance testing.

4.6.2.5 Speed. A measured distance of 44 feet shall be laid out over a dry, smooth, level, concrete surface. The beginning and end of the test course shall be clearly identified. The loaded truck shall be operated a sufficient distance to attain maximum speed prior to entering the measured test course. With a timing device, record the time required for truck to travel the measured distance. Drive truck in the reverse direction and record travel time over the same measured distance. This procedure shall be repeated at least 3 times. Alternate methods for checking speed may be used, such as electric eyes, electronic or other accurate means, and when used, may be timed through shorter distance. The time required for travel in each direction of travel shall be separately averaged. The averaged time in each direction shall not exceed 7.5 seconds. Test is to be conducted with a fully charged battery and with tires warmed-up by 10 or 15 minutes of running.

4.6.2.6 Acceleration. A measured distance of 22 feet shall be laid out over a dry, smooth, level, concrete surface. A reference point shall be marked on the truck so as to be readily seen by the operator and test personnel while truck is moving. The truck with rated load shall be positioned so that the reference mark on the truck is directly over the starting mark. The truck shall be accelerated as in normal operation and the time recorded when the truck reference mark moves from beginning to end of test distance. This shall be repeated not less than three times in both the forward and reverse directions. The time required to travel in each direction shall be separately averaged. Time for travel in each direction shall not exceed the time specified in 3.6.7.

4.6.2.6.1 Slope ascension, forward direction. The truck with rated load in carrying position shall be driven onto a 15 foot long ramp having a 10 percent slope. Truck shall be brought to a complete stop when all wheels are positioned on the ramp. Accelerate truck from this location up the ramp to a horizontal level. Test shall be conducted with a fully charged battery.

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4.6.2.7 Maneuverability. The test shall be conducted in a 78 or 81 inch aisle, as applicable (see 3.6.16), with side panels or enclosures at least 5 feet high and a stacking base for load placement not less than 7 feet high. The stacking base shall be framed or enclosed with side panels to simulate a typical storage rack 56 inches wide for style I trucks and 54 inches wide for style II trucks. Test load shall be a 40-inch by 48-inch pallet with rated load. Time measurements for each placement, or removal cycle shall begin with the truck parallel and within the aisle with forks approximately 8 inches above the floor in the retracted position, and the cycle shall end with the truck in approximately the same position (forks empty or loaded) in the lowered position. Time measurements shall include the time required to maneuver, to stack, lifting load to stack height, extending load, placement on the stacking base, return with forks empty, and lowering forks to 8 inches lift height. Similarly, the reverse cycle shall include lifting empty forks to the height of the pallet on the stacking base, maneuvering into position, engagin the pallet, raising the load off the stacking base, retracting load, returning to the original position, and lowering load to carrying position. The operation shall be perfomred at least 9 out of 10 times, without reversal of direction, with not more than 60 seconds being required for each placement or removal cycle. This testing may be conducted by contractor's selected operator, utilizing such aids and skillful manipulation as may be required to provide the best mechanical advantage to the opeator and the equipment. Operation may include, but not be limited to: (1) maneuvering with load raised to an efficient height, but not higher than the stacking base for the pallet; (2) operation with load in rearward tilt position; (3) in the removal of a pallet from stack position, entry of the forks into the pallet opening prior to attaining a position for the truck in which its longitudinal axis is fully perpendicular to the face line of the pallet stack; (4) similarly, but conversely, in the removal of the truck forks from a pallet which has been placed in position, truck motion may follow any preferred pattern of movement to permit most efficient withdrawal possible without damaging adjacent or opposite pallet stack or barriers. In placement and load removal cycles, the pallet and load support on the forks shall be with any vertical face or side at 90 degrees to either the longitudinal or lateral axis. The operator may use any other recognized techniques or manipulation commonly practiced in commercial or military handling operations, with the exception of direct reversal during any one placement or removal action.

4.6.2.8 Load back rest. A static thrust load equivalent to rated load of the truck shall be applied to the fork rack uniformly distributed over the entire vertical surface from the load supporting surface of the forks (forks retracted) to the extreme top of the load backrest without causing fracture at the welded joints, deformation of any frame member or failure in the rack mounting.

4.6.2.9 Lift assembly.

4.6.2.9.1 Shock. The lift mechanism shall be tested for effectiveness to absorb shock and prevent overturning of the truck and damage to lift system, when rated load is lowered at maximum speed in retracted position and fully actuated control is released abruptly. Repeat test 20 times and examine truck to determine compliance with specified requirements.

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4.6.2.9.2 Retraction Overload. With pressure relief valve inoperative, raise retracted forks to a minimum of 30 inches lift height and place 300 percent of rated load on forks and retain for ten minutes. Remove load and examine truck for evidence of external leaks or breakage of the hydraulic system as a result of this test. The load support assembly, frame, and related structure shall be examined for permanent deformation or any indication of structural failure.

4.6.2.9.3 Extension overload. Extend load support assembly to full forward position, with forks raised at least 12 inches above the floor and rear of truck counterweighted or tied down. Place 200 percent rated load on the fork assembly and retain for 5 minutes. Remove load and examine the truck for evidence of failures. The pressure relief valve shall be made inoperative for this test.

4.6.2.10 Stability. Tests to determine compliance with the stability requirements of 3.6.17 shall be conducted in accordance with ANSI B56.1.

4.6.2.11 Frame racking test. Determine maximum static force for each support under loaded and unloaded conditions with forks extended and retracted. Secure all but one support area of the truck in a jig or similar device, and apply a vertical force equal to 200 percent of the static force determined for each support area. Complete this test by applying load first at one and then the other of two adjacent corners. Examine frame and related structure for permanent deformation or distortion as a result of this test.

4.6.2.12 Overhead safety guard. The overhead safety guard shall be tested in accordance with ANSI B56.1. Nonconformance to the requirements of 3.6.18 shall constitute failure of this test.

4.6.2.13 Operator's platform. With the truck anchored to prevent movement apply a force equal to the loaded weight of the truck. The load shall be applied to an area 3-1/2 inches high and 3 inches wide maximum with its center 42 inches above the floor. The load shall be applied at the right hand rear corner of the compartment. Any evidence of failure or permanent deformation shall constitute failure of this test.

4.6.2.14 Service brake test. Place truck on a level surface, with rated load positioned on the forks with center of gravity of load placed 24 inches from front face of forks. The forks shall be fully retracted and load raised to the load carrying position of 17-24 inches above the floor. With brakes released, apply a horizontal drawbar force in the direction of the longitudinal centerline of the truck at the height of the center of gravity of the truck and the rated load combined. After the truck is in motion, allow the brake to be applied and read the drawbar drag on a dynamometer. Perform the test in the reverse direction and measure the drawbar drag. Failure to conform to 3.6.8 shall be cause for rejection of the truck.

4.6.2.15 Power train static test. The static torque test shall be accomplished by using a torque arm with one end attached to the motor shaft and with the other end attached to a dynamometer. Drive wheel(s) shall be locked to prevent any movement. The force shall be applied through the dynamometer to the torque arm by use of chainfall or similar device. Applied torque shall be equivalent to 150 percent of the maximum torque cable of being developed by the

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traction motor. At the completion of the test, disassemble and inspect the power train for possible failure due to external and internal stresses. This test may be accomplished on the entire power train assembly prior to assembly of the truck, at the option of the contractor.

4.6.2.16 Electromagnetic interference. The preproduction model equipped for control of electromagnetic interference in accordance with 3.10 shall be tested as specified in MIL-STD-461. The supplier shall furnish the contracting officer the report required by MIL-STD-461. Upon approval of the report by the contracting officer and provided all other requirements of the specification are met, the preproduction model shall be used as a model for all production trucks.

4.6.2.17 Power consumption. The truck, equipped with a fully charged battery as specified in 3.2.2.1, shall be operated continuously over the test course shown on figure 1. A single fully charged battery shall be used for the duration of the test. Truck operation shall not be stopped except to alternate operators. The truck shall be operated in low speed so as to complete each test cycle in 3 minutes and 30 seconds plus or minus 15 seconds. Each test cycle shall be timed. Test course shall be in accordance with figure 1. Test loads weighing 3,000 Pounds, plus zero, minus 30 pounds, shall be placed as indicated. Test shall be conducted commencing at point "A". Pick up load at "A", move to point "C", extend load and place on floor. Back truck to point "B", pick up load and move to point "C". Raise and extend load and place on top of load "A". Retract forks, lower forks to floor, raise and extend forks. Pick up load and move to point "B". Place load at point "B" and back truck to point "C". Pick up load and move to point "A". Place load at point "A" and retract forks to complete cycle. Repeat cycle for a minimum of 5 hours. Inability to operate for a minimum of 5 hours maintaining cycle times as specified shall constitute failure of this test.

4.6.2.18 Endurance. The truck shall be subjected to 2,000 complete circuits of the test course shown in figure 2. The test shall be conducted on a continuous basis during each operating 8 hour day. Battery changes shall be permitted as necesary to assure continuous operation for each operating day. At the completion of each 1,000 circuits of the course, manufacturer's recommended preventive maintenance shall be performed and the truck inspected to assure all components are functioning properly. No repair or replacement of parts shall be permitted. The course shall be negotiated one-half the total circuits in a clockwise direction and one-half in a counterclockwise direction as follows. The course need not be rearranged for the counterclockwise operation. Tests load weighing 3000 pounds, plus zero, minus 30 pounds, may be placed on standard 40-inch by 48-inch pallets. From point "A" proceed to point "B" and remove load from platform. Retract load fully and proceed rearward to point "C" with load at normal carry height. Deposit load on the platform at point "C", retract forks and proceed to point "D". Pick up load at point "D" and proceed rearward to starting point "A". At point "A" stop the truck, extend the forks and lower to ground level, blow horn, actuate light switch, and fully apply brake. Raise and retract forks to carry position. Proceed forward to "B" and deposit load. Continue to negotiate the course in this manner, alternately placing and retrieving loads until the required number of test circuits has been completed. All turns into and out of the stack areas shall be smooth and continuous. Inability to negotiate the course for 2,000 complete cycles, evidence of excessive wear, or failure of any component shall

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constitute failure of this test.

4.6.2.19 Battery removal. Measure force to maintain movement of battery to determine conformance to requirement of 3.2.2.2.

4.7 Post-test inspection. Upon completion of all tests specified herein, each operating part and related structure and integral part of the sample truck shall be examined in sufficient detail for presence of any defects or excessive wear which caused failure during tests or might cause failure in subsequent operation of trucks.

4.8 Inspection comparison. The Government may select one or more trucks at any time during the contract production period and subject these trucks to the examination specified in 4.6.1 and the tests specified 4.6.2.1 through 4.6.2.19 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on trucks selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production trucks.

In addition to any test specified as part of the inspection comparison, the Government reserve the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.8.1 Inspection failure. Failure of an inspection comparison truck to meet any requirement specified herein during and as a result of the examination and tests specified in 4.8 shall be cause for rejection of the inspection comparison truck and shall be cause for refusal by the Government to continue acceptance of production trucks until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government or trucks previously accepted and produced under the contract. Any deficiencies found as a result of the inspection compariosn will be considered prima facie evidence that all trucks accepted prior to the completion of the inspection comparison are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.9 Inspection of preparation for delivery. The trucks shall be examined to determine conformance to the preservation, packaging, packing and marking requirements of MIL-STD-162.

4.10 Certification. A certification of compliance or test data showing compliance with transportability requirements of this specification (3.16) shall be provided within 60 days after contract award for Army equipment. Send certificate of compliance with transportability requirements and supporting data to Commander, Military Traffic Management Command, Attn: MT-SA, Washington, DC 20315, and an information copy to Commander, Military Traffic Management Command Transportation Engineering Agency, P. O. Box 6276, Newport News, VA 23606.

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Preservation, packaging and packing shall be level A, B, or C as specified, (see 6.2), in accordance with MIL-STD-162.

5.2 Marking. In addition to any special marking specified in the contract, marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Trucks described herein are intended primarily for stacking and unstacking palletized cargo into or from pallet racks. They are also intended for handling palletized cargo in warehouses, where narrow aisle widths and low floor load ratings prevent the use of less maneuverable and heavier equipment. Trucks are intended for operation over paved surfaces for distances of approximately 100 feet per round trip.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Style and size required. (see 1.2).

(c) When preproduction model is not required. (See 3.1.1).

(d) Time frame required for submission of preproduction model. (See 3.1.1).

(e) When the Government will conduct any or all of the preproduction model examination and tests. (See 3.1.1).

(f) Which examination and tests to be conducted by the Government and which examination and tests to be conducted by the supplier. (See 3.1.1).

(g) When initial production inspection is required and number of trucks to be furnished. (See 3.1.6).

(h) When batteries will not be furnished by the Supplier. (See 3.2.2.1).

(i) When batteries shall be other than as specified. (See 3.2.2.1).

(j) When fork dimensions are to be other than shown in Table I. (See 3.3.3.1).

(k) When certificate of compliance shall be furnished in lieu of drive line test. (See 3.2.3).

(l) When frame racking test is not required. (See 3.3.1).

(m) When remote control is required. (See 3.4.1.5).

(n) Electromagnetic interference requirements. (See 3.10).

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(o) Level of preservation, packaging and packing required. (See 5.1).

(p) Classification of preservation, packaging and packing required. (See 5.1)

(q) Specify transportability modes (See 3.15.7).

(r) When fixed overhead guard is not required (See 3.5.2)

6.3 Preproduction model. Any changes or deviations of production trucks from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish trucks conforming to this specification.

6.4 Lubricants and oils. The contractor shall furnish a list of military lubricants applicable to the truck covered by this specification.

6.5 Noise. When the noise level of the forklift is 85 dB(A) or greater the appropriate discussion of the noise hazard per MIL-STD-1474 (MANUALS Paragraph) should include the requirement for hearing protection, the oise level of the forklift, the distance from the forklift at which 85dB(A) will always be met, and what operational situations or system configuration will reduce the level to 85dB(A).

6.6 Incident report. When the supplier conducts the tests specified herein, a verbal or teletype notification within 24 hours and a written report within 48 hours shall be furnished the contracting officer of any incident of equipment malfunction or failure during the conduct of the test. As a minimum, the report shall describe components and parts affected, test and operatig conditions, date of incident, hour meter reading, how detected, and description of incident.

Custodians:
Army - ME
Navy - SA
Air Force - 99

Review Activities

Army - MT, AT
Navy - YD
DLA - CS

Preparing activity:
Navy - SA

Project 3930-0510

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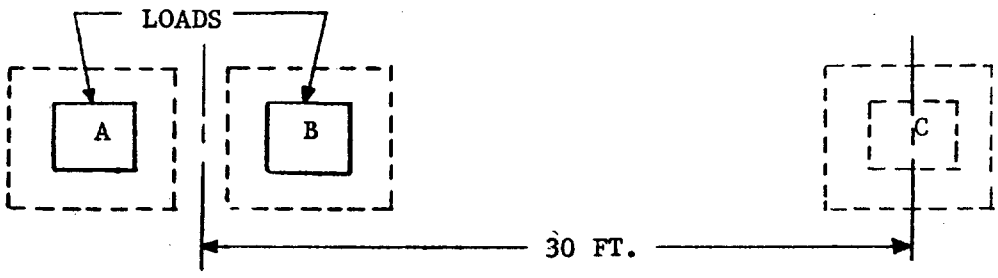
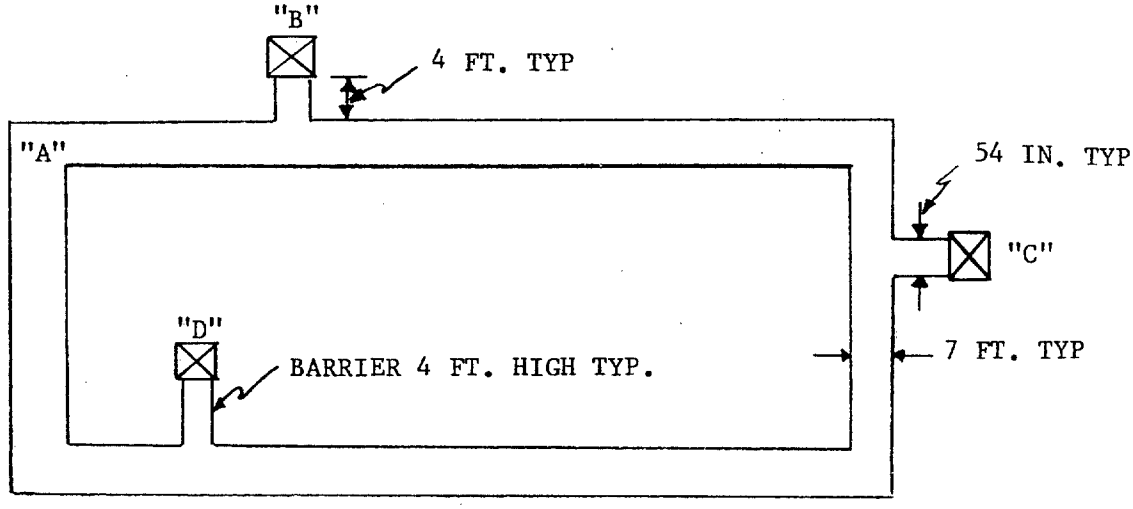



FIGURE 1. POWER CONSUMPTION



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| POINT | STACK HEIGHT 130"L.H. | 144"L.H. | 180"L.H. | 198"L.H. |
|---|---|---|---|---|
| B | 124" | 138" | 174" | 192" |
| C | 60" | 70" | 90" | 100" |
| D | FLOOR | FLOOR | FLOOR | FLOOR |




TOTAL COURSE LENGTH - 320 FT. APPOX.

FIGURE 2. ENDURANCE COURSE

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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-T-21643C

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED S/N 0102-014-1802

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